How posts spread on X

Week 3, Lecture 1 · Grow on X: From Zero to 10,000 Followers

Autumn 2026

What we'll cover

- The three-stage pipeline: candidate sourcing, light ranker, heavy ranker
- Signal weights: replies at 13.5x, retweets at 20x, likes at 1x baseline
- Engagement velocity: why the first 30 minutes determine total reach
- Time decay and the second-wave effect
- Why dwell time and "show more" clicks matter more than likes

By the end of this lecture you can score any post by its likely algorithmic performance before you publish it.

The three-stage recommendation pipeline

Candidate sourcing

- From 500M daily posts, the algorithm pulls roughly 1,500 candidates for each user's For You feed (X Engineering 2023)
- Two pools: posts from accounts you follow, posts from accounts you do not
- The 50/50 split: about half your For You feed comes from unfollowed accounts (Hurler, Gizmodo 2023)
- Sourcing is where most posts are eliminated: they never enter the ranking stages at all

Getting into the candidate pool requires meeting a minimum threshold of engagement in the first minutes after posting.

Light ranker and heavy ranker

- Light ranker: fast, cheap filters. Removes obviously irrelevant or low-quality candidates.
- Heavy ranker: a neural network that scores remaining candidates on predicted engagement probability (X Engineering 2023)
- The heavy ranker assigns probability labels across multiple signal types: explicit (likes, replies, reposts) and implicit (profile clicks, tweet clicks, dwell time)
- "The algorithm takes into account thousands of features to assign engagement probability labels." (Hurler 2023)

The heavy ranker's output determines the order of posts in your For You feed.

The pipeline in one picture

```
500M daily posts
      |
 Candidate sourcing (~1,500 per user)
      |
 Light ranker (fast filters)
      |
 Heavy ranker (neural network, engagement probability)
      |
 For You feed
```

Your post must survive all three stages. Most do not reach stage two.

Signal weights

What the algorithm actually values

Weights from Hashmeta (2025), based on the open-sourced algorithm:

Signal	Weight
Retweet	20x
Quote tweet	15x
Reply	13.5x
Like	1x (baseline)

One retweet is worth as much as 20 likes. One reply is worth 13.5 likes.

Implicit signals matter as much as explicit ones

Explicit signals (likes, replies, reposts) are visible to everyone. Implicit signals are harder to game:

- **Profile click:** the reader clicked your name after seeing the post
- **Tweet click / show more:** the reader expanded the post to read the full text
- **Dwell time:** the reader stopped scrolling and spent time on the post

Dwell time and "show more" clicks are quality signals the algorithm uses precisely because they cannot be faked at scale.

What the algorithm penalizes

- External links: reach penalty of 30-50% (Hashmeta 2025)
- Mentions of out-of-network URLs and competitor platforms get deboosted (Hutchinson, Social Media Today 2023)
- X is testing an in-app browser to keep users on-platform and preserve dwell-time signals (Hutchinson 2025)

Practical rule: put the link in the first reply, not the post body. The post signals quality; the link travels separately.

Engagement velocity

The first 30 minutes determine total reach

- When you publish, the algorithm shows your post to a small test sample
- If that sample engages (replies, reposts), the post is promoted to a larger batch
- If it does not, the post is demoted and most of your followers never see it
- This feedback loop runs in minutes, not hours

The first 30 minutes are not the start of distribution. They are the distribution decision.

What early velocity actually requires

- A hook strong enough that people reply within the first few minutes
- An audience that is online when you post: 9pm Tuesday and Wednesday show the highest connected-user density (Metricool 2024 study of 23,561 accounts)
- Replies from your own network: responding to your own replies in the first 30 minutes keeps the thread active and signals conversation quality
- One repost from a peer (20x weight) outweighs 20 early likes

Velocity is a product of timing, hook quality, and your existing network behavior.

Premium vs. free: the initial reach asymmetry

- Premium (Blue) accounts receive 40-80% initial reach to followers in the first push
- Free accounts receive 10-20% (Hashmeta 2025)
- This means a free account with 1,000 followers may reach only 100-200 people in the first wave
- For free accounts, early replies from existing followers are even more critical: they compensate for the smaller initial distribution

You cannot buy velocity, but you can structure your posting time and reply behavior to maximize it.

Time decay and the second wave

How posts age

- The heavy ranker applies a time-decay penalty as a post ages
- Most posts peak within two hours and go quiet within six
- But some posts re-enter the For You feed in a "second wave" 12-48 hours later
- Second waves happen when a post gains a high-weight signal (a repost from a large account) after the initial velocity window closes

Second waves are not predictable. The condition that triggers them is reaching a new node in the network: an account whose followers have not seen the post.

Designing for the second wave

- A post with high dwell time and a good "show more" rate builds a quality score that persists
- If a large account reposts hours later, the algorithm treats it as a fresh signal
- Thread lead-in tweets are better candidates for second waves because readers share the thread link, not individual tweets
- Chen (2023): "Creator traffic is driven by social feed algos, which lends itself to big spikes in traffic that appear and then go away." The second wave is the second spike.

You cannot manufacture a second wave. You can write posts that deserve one.

Take-aways

1. The X recommendation pipeline has three stages: candidate sourcing, light ranker, heavy ranker. Most posts are eliminated at sourcing.
2. Signal weights (Hashmeta 2025): retweets 20x, quotes 15x, replies 13.5x, likes 1x. Design posts that earn the high-weight signals.
3. Engagement velocity in the first 30 minutes determines total reach. Post when your audience is online.
4. Dwell time and "show more" clicks are implicit quality signals that cannot be gamed. They reward genuinely interesting content.
5. External links carry a 30-50% reach penalty. Move links to the first reply.

> "The algorithm takes into account thousands of features to assign engagement probability labels."

Kevin Hurler, *Twitter's Recommendation Algorithm Is Now Open Source*, Gizmodo (2023)

What's next

- **Section (this week):** Reply sprint. Ten accounts, ten replies, five DMs, track profile clicks.
- **Reading:** Week 3 reading at `/c/grow-on-x-26au/readings/wk03` (the X algorithm and the reply game)
- **Lecture 2:** The reply game and reaching out first
- **HW2 (twenty posts):** due this week